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Analyzing Factors Influencing the Popularity of Computer Monitors in Online Markets

Introduction

This project analyzes factors influencing the popularity of computer monitors in online markets (Amazon).
Popularity is approximated using review count, which serves as a proxy for sales volume. It is analyzed together with several key factors, including product features (resolution, screen size), price and market demand (Google Trends).
The goal is to identify which factors have the strongest impact on popularity and pricing.

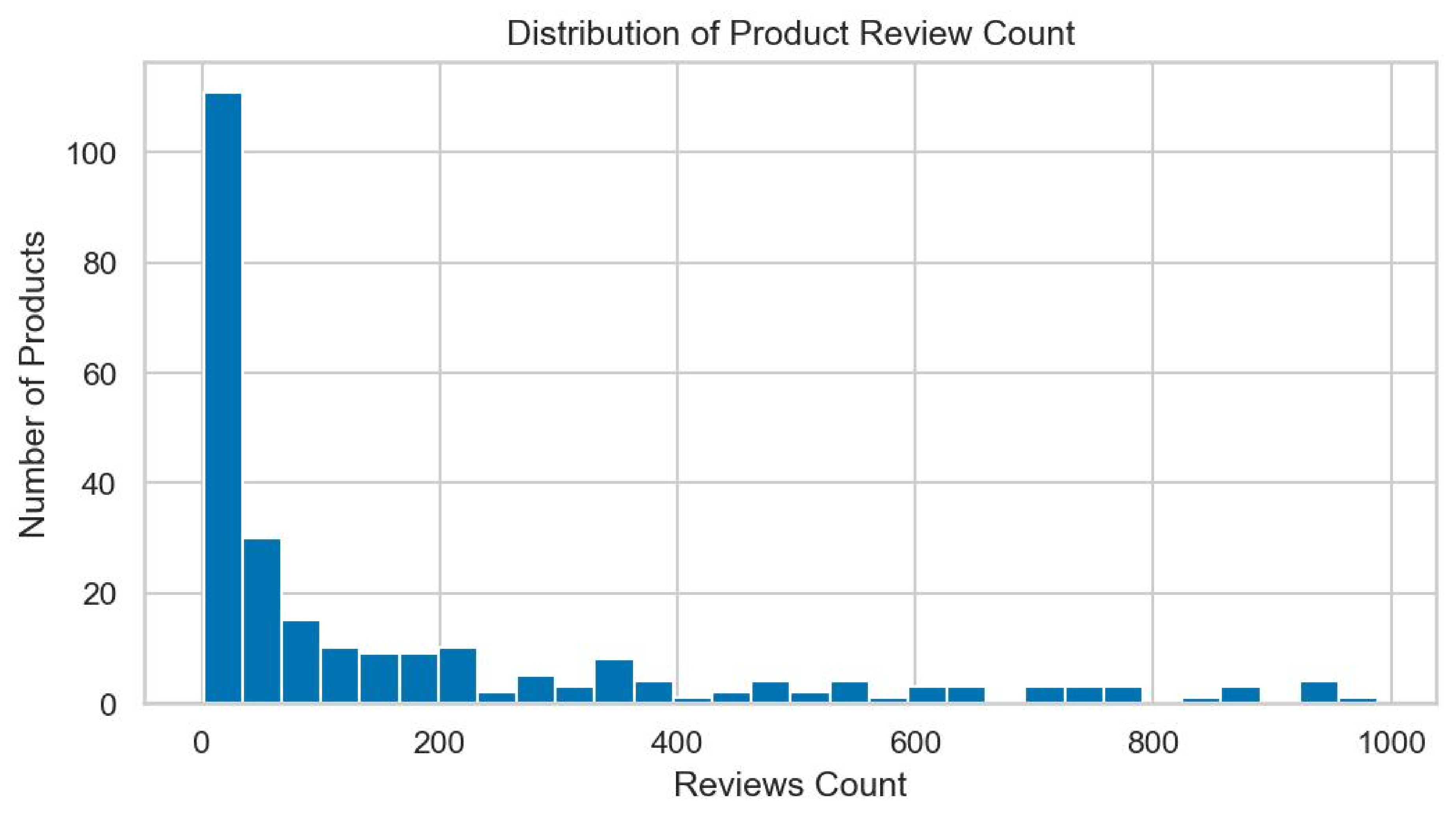
Data Sources

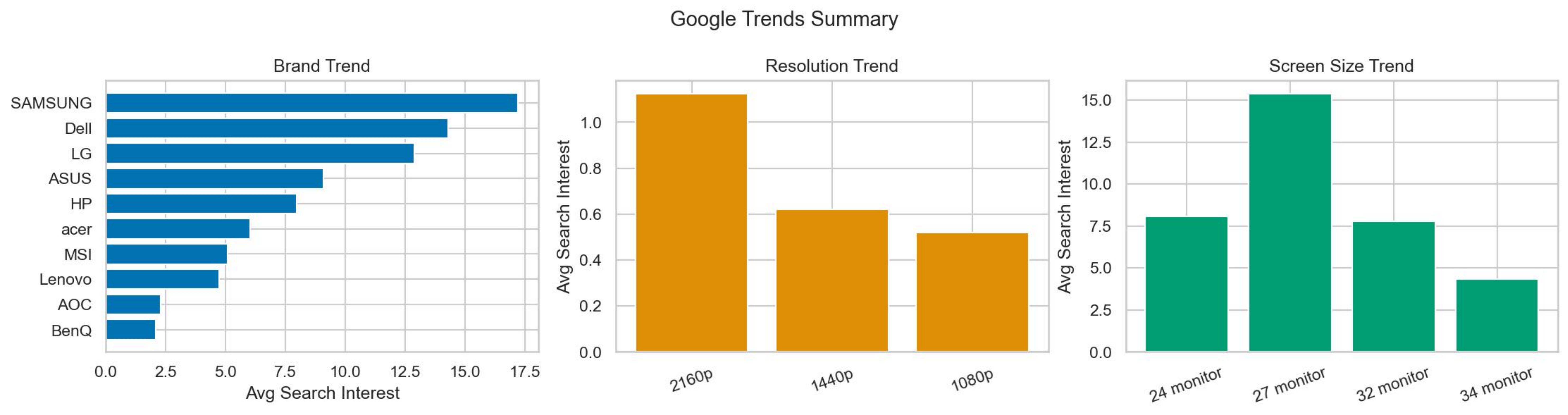
Data Souce	Description	Data Collection & Processing	Data Size
Amazon Monitor Dataset (Kaggle)	Product-level data including monitor specifications such as title, price, resolution, screen size, and brand information (collected in 2022).	- collected via API into CSV - cleaned titles - standardized features	270 unique products
Google Trends (pytrends API)	Search interest data for monitor-related keywords, including brands, resolution (4K, QHD, FHD) and screen size (24–34 inch).	- collected via pytrends - averaged time-series data	34 keywords
Amazon Review Data (Web scraping)	Product-level review information including review count (ratings_total) and rating scores.	- used Playwright to simulate real browser sessions on Amazon - removed outliers	270 products

Results

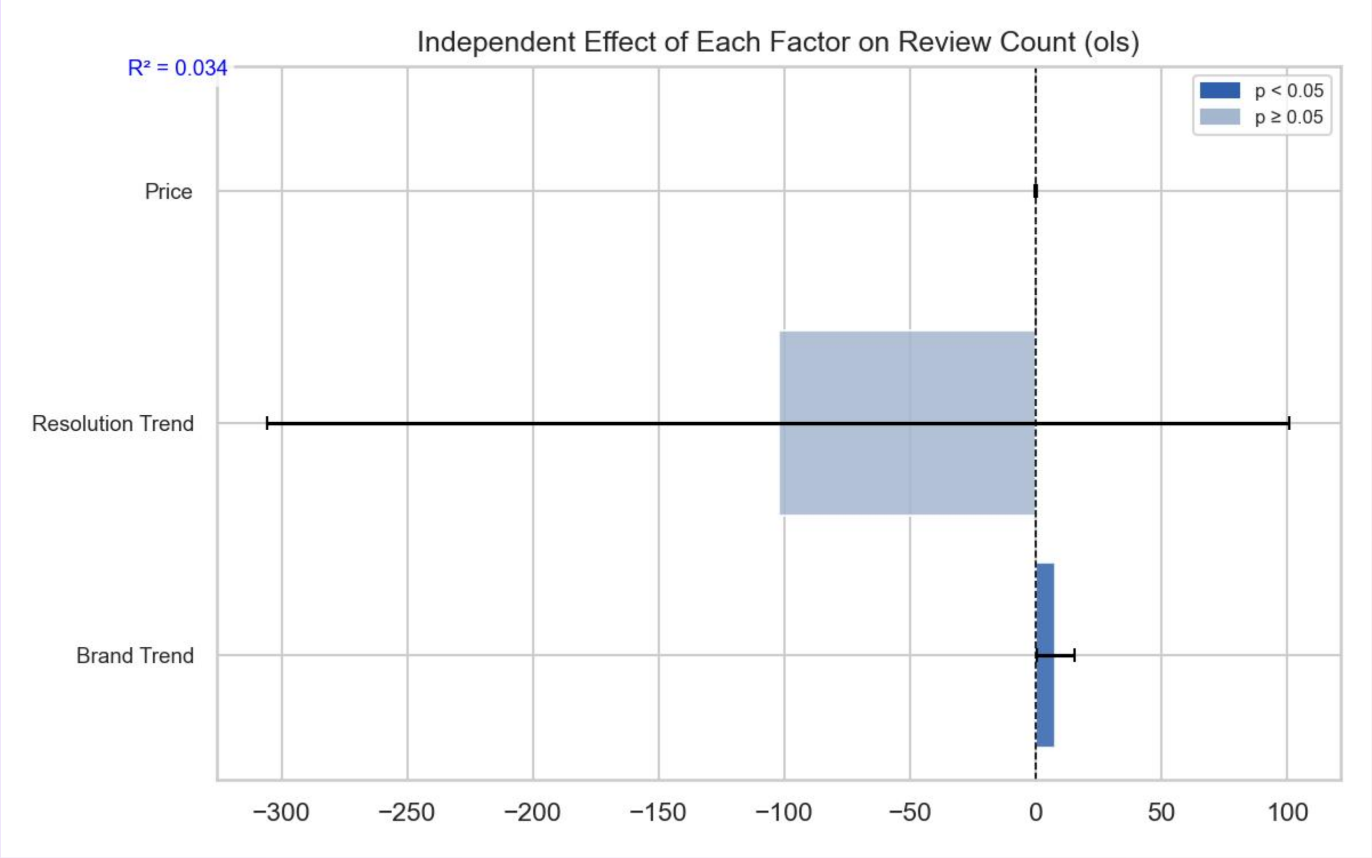
Review Count Distribution:
- Review count is highly skewed, with a small number of products receiving the majority of reviews
- Most products have relatively low review counts, indicating uneven popularity distribution.

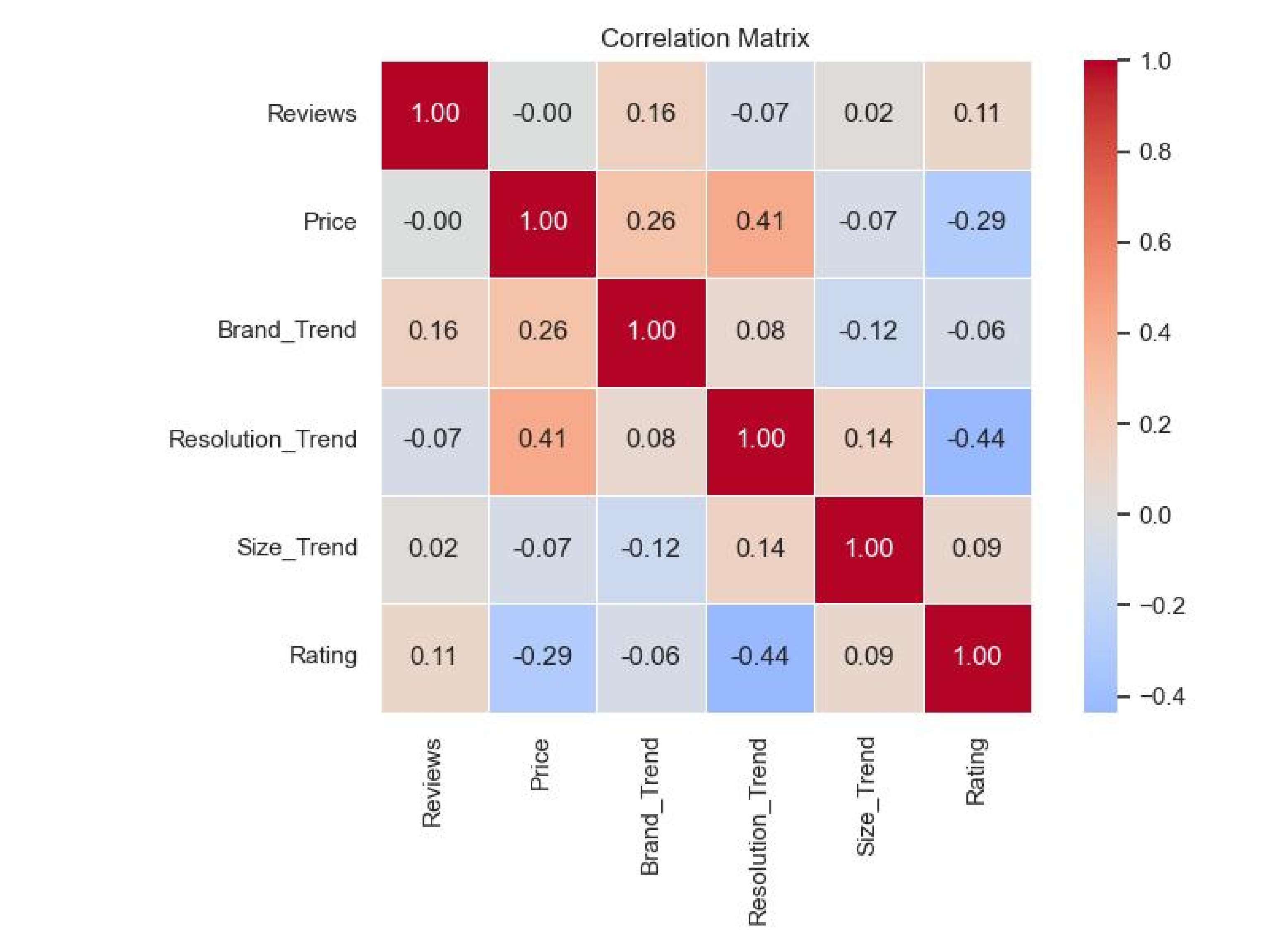
Market demand:
- Brand search interest varies significantly across companies
- Samsung and Dell show the highest search interest
- Higher resolution monitors (4K) have the strongest demand

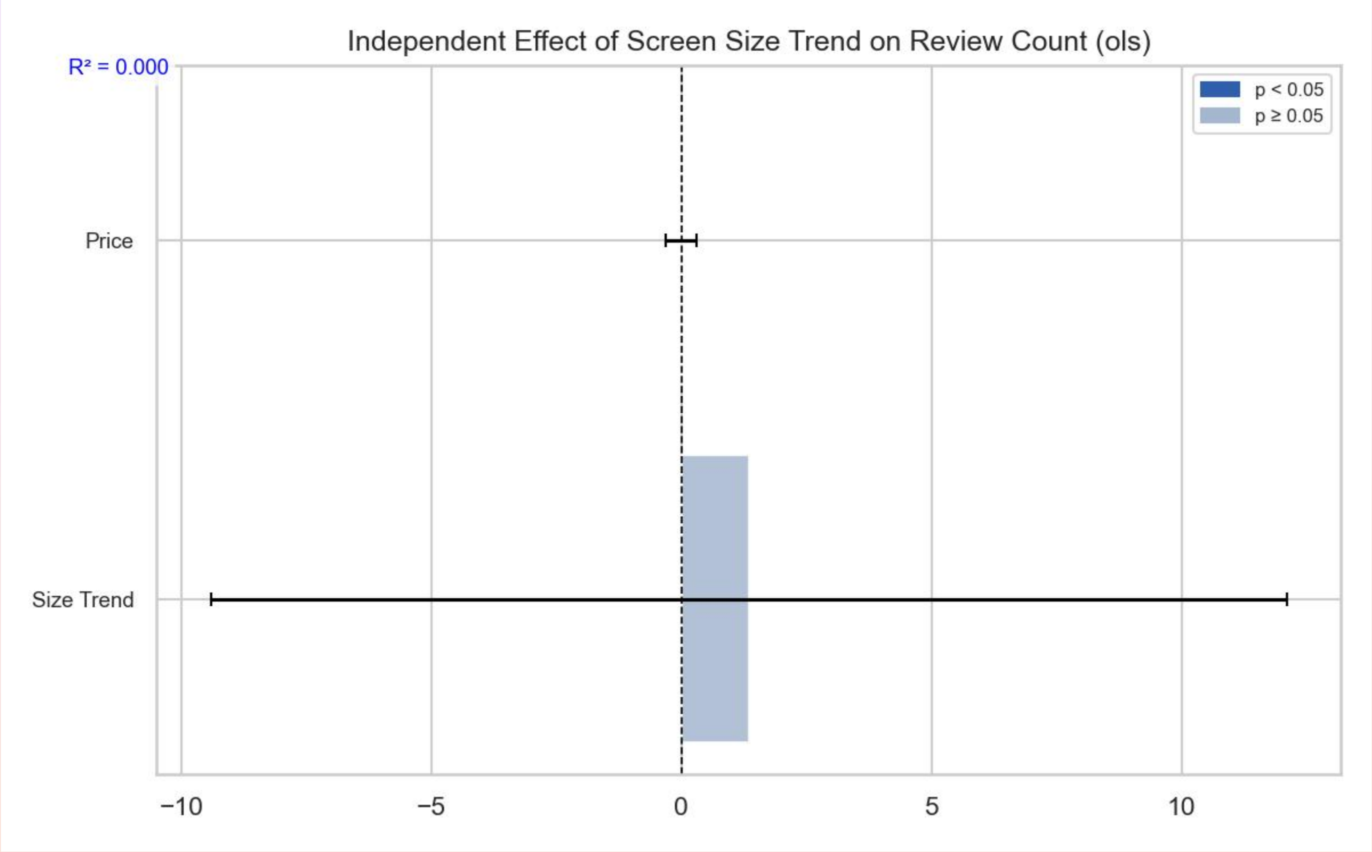




Results



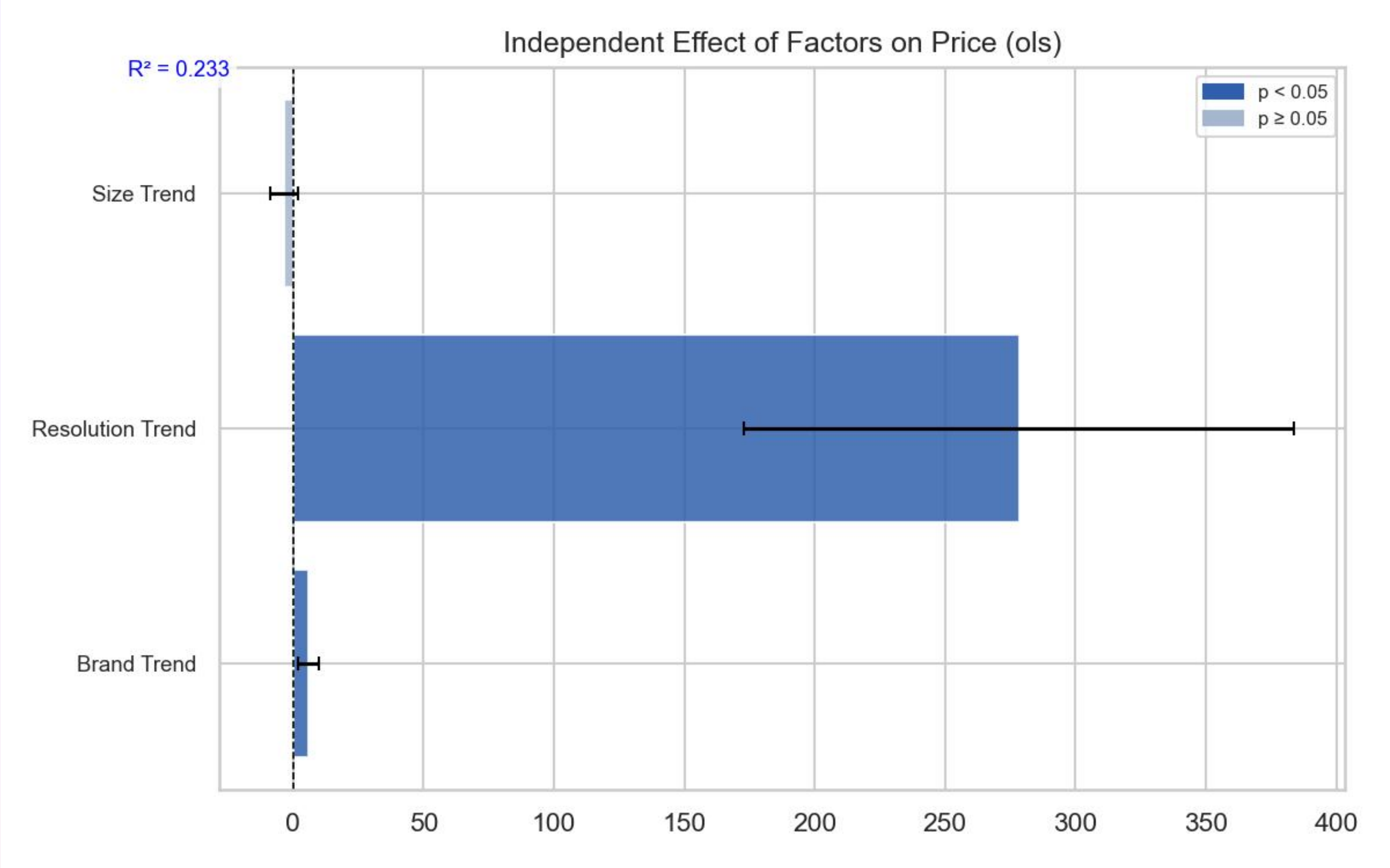


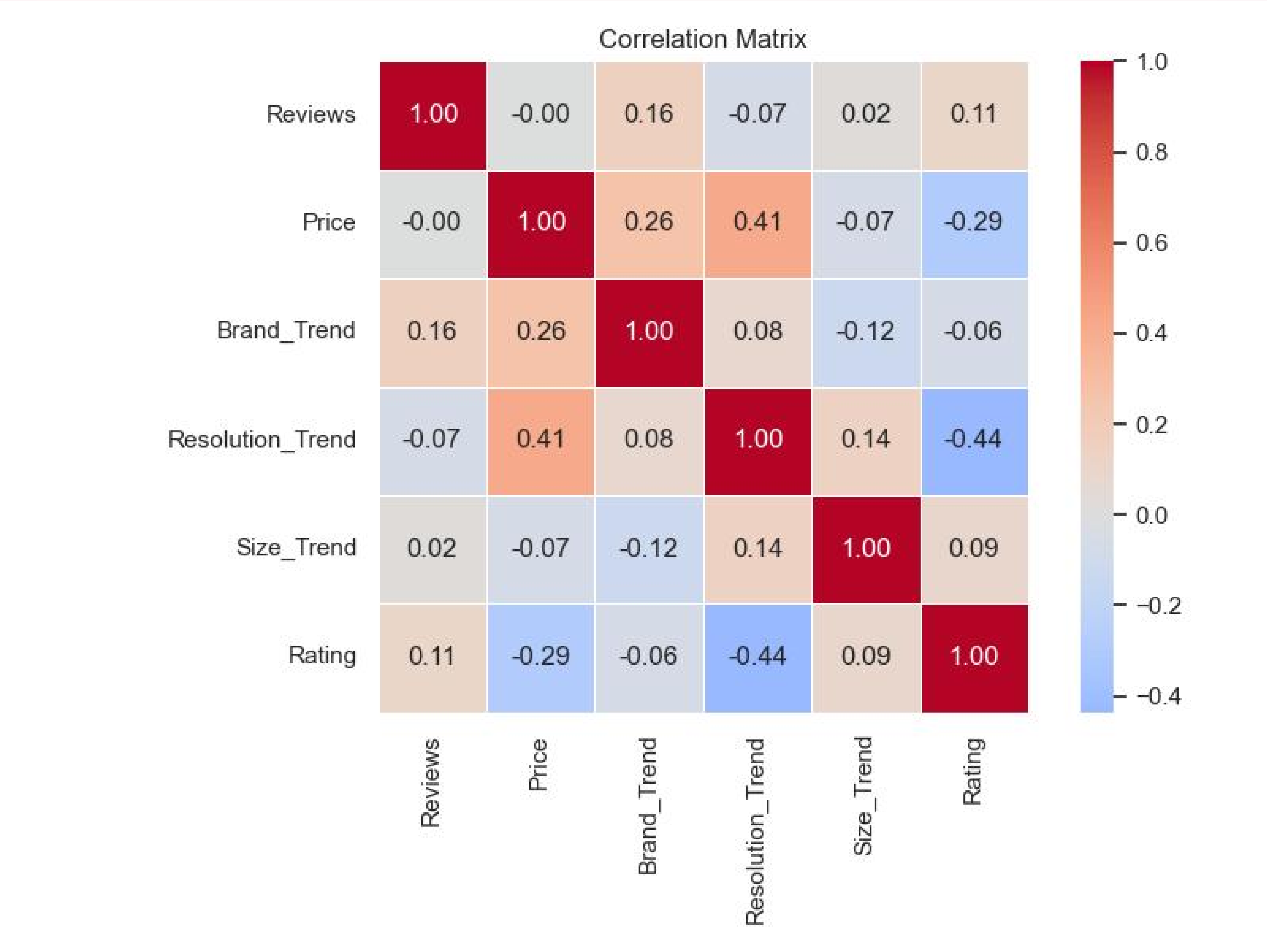


- Correlation analysis reveals weak relationships between all variables and review count.
- The strongest correlation is between Price and Resolution Trend (r = 0.41)
- Review count shows low correlation with all features (max r = 0.16)

- Brand Trend shows a significant positive effect on review count
- Resolution Trend is not statistically significant (wide confidence interval crossing zero)
- Screen Size Trend is not statistically significant
- Price remains weakly related to popularity

Results





- Correlation analysis shows the strongest relationships are associated with price

- A second regression model is built to examine factors influencing price. Results show:
(1) Resolution Trend has a strong positive and significant effect on price
(2) Brand Trend has a small positive effect, but not statistically significant effect
(3) Screen Size Trend is not significant

Conclusion

- Popularity is highly uneven, with a small fraction of "superstar" products capturing the vast majority of consumer feedback.
- Market demand (Brand Trend) is the most consistent predictor of popularity, despite weak surface correlations.
- Resolution Trend is the primary driver of Price.
- Screen Size Trend failed to reach statistical significance across multiple models, showing little independent influence on either Price or Review Count.

Challenges

- Difficulty in aggregating Google Trends into a single score due to multiple possible keywords

- The most difficult part was data collection. For example, Amazon actively blocks automated scraping, requiring Playwright browser simulation and randomized delays.

- Only ~193 usable records after cleaning, reducing statistical power and contributing to weak model performance

Thank You.